Book: Measurement and Data Analysis for Engineering and Science by Patrick Dunn

in which τ denotes an arbitrary time measured from time t_1. Both equations represent **ensemble averages**. This is because $\mu_x(t_1)$ and $R_x(t_1, t_1 + \tau)$ are determined by performing averages over the ensemble at time t_1.

If the values of $\mu_x(t_1)$ and $R_x(t_1, t_1 + \tau)$ change with t_1, then the signal is **nonstationary**. Otherwise, it is **stationary** (stationary in the *wide* sense). A nondeterministic signal is considered to be **weakly stationary** when only $\mu_x(t_1) = \mu_x$ and $R_x(t_1, t_1 + \tau) = R_x(\tau)$, that is, when only the signal's ensemble mean and autocorrelation function are time invariant. In a more restrictive sense, if all other ensemble higher-order moments and joint moments (see Chapter 7 for more about moments) also are time invariant, the signal is **strongly stationary** (stationary in the *strict* sense). So, the term stationary means that each of a signal's ensemble-averaged statistical properties are constant with respect to t_1. It does *not* mean that the amplitude of the signal is constant over time. In fact, a random signal is never stationary in time!

For a single time history, the temporal mean value, μ_x, and the temporal autocorrelation coefficient, $R_x(\tau)$, are

$$\mu_x = \lim_{T\to\infty} \frac{1}{T} \int_0^T x(t)\,dt \qquad \textbf{[11.4]}$$

and

$$R_x(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_0^T x(t)x(t+\tau)\,dt. \qquad \textbf{[11.5]}$$

For most stationary data, the ensemble averages at an arbitrary time, t_1, will equal their corresponding temporal averages computed for an arbitrary single time history in the ensemble. When this is true, the signal is **ergodic**. If the signal is periodic, then the limits in Equations 11.4 and 11.5 do not exist because averaging over one time period is sufficient. Ergodic signals are important because all of their properties can be determined by performing time averages over a *single* time history record. This greatly simplifies data acquisition and reduction. Most random signals representing stationary physical phenomena are ergodic.

A finite record of data of an ergodic random process can be used in conjunction with probabilistic methods to quantify the statistical properties of an underlying process. For example, it can be used to determine a random variable's true mean value within a certain confidence limit. These methods also can be applied to deterministic signals, which are considered next.

11.4 SIGNAL VARIABLES

Most waveforms can be written in terms of sums of sines and cosines, as will be shown later in Section 11.7. Before examining more complex waveform expressions, the variables involved in simple waveform expressions must be defined. This can be done by examining the following expression for a simple, periodic sine function:

$$y(t) = C\sin(n\omega t + \phi) = C\sin(2\pi nft + \phi), \qquad [11.6]$$

in which the argument of the sine is in units of radians. The variables and their units are as follows:

- C: amplitude [units of $y(t)$]
- n: number of cycles [dimensionless]
- ω: *circular* frequency [rad/s]
- f: *cyclic* frequency [cycles/s = Hz]
- t: time [s]
- T: period ($= 2\pi/\omega = 1/f$) [s/cycle]
- ϕ: phase [rad] where $\phi = 2\pi(t/T) = 2\pi(\theta°/360°)$

Also note that 2π rad = 1 cycle = 360° and $\sin(\omega t + \pi/2) = \cos(\omega t)$. The top plot in Figure 11.4 displays the signal $y(t) = \sin(\pi t)$. Its period equals $2\pi/\pi = 2$ s, as seen in the plot. The preceding definitions can be applied readily to determine the frequencies of a periodic signal, as in Example 11.1.

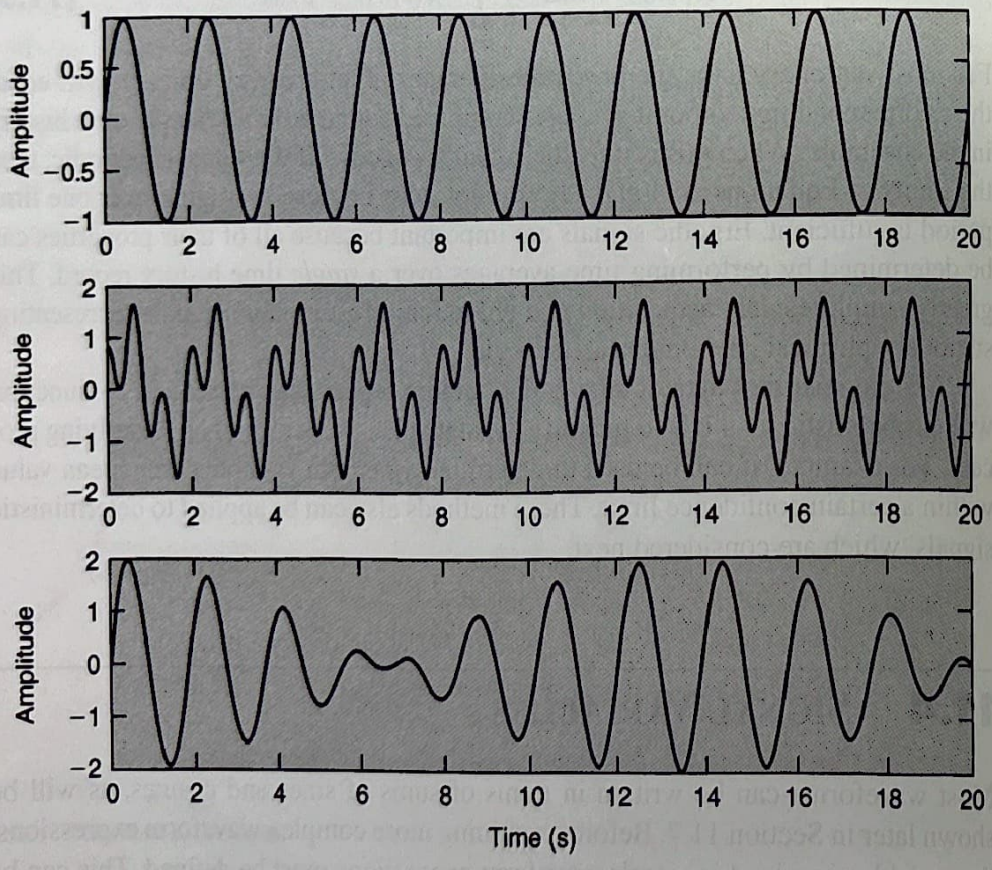


Figure 11.4 Various signals comprised of sines and cosines.

11.1 MATLAB SIDEBAR

MATLAB has a number of built-in functions that generate standard signals, such as square, triangular, and sawtooth waves, and the trigonometric functions. A square wave with a frequency of f Hz and an amplitude that varies from $-A$ to $+A$ over the time period from 0 to 7 s is generated and plotted using the MATLAB command sequence

```
t = 0:0.001:7;
sq = A*square(2*pi*f*t);
plot(t,sq)
```

The MATLAB `sawtooth(t,width)` function produces a sawtooth wave with period 2*pi. The fraction of the period at which sawtooth's peak occurs is specified its width argument, whose value varies between 0 and 1. A triangular wave is produced when width equals 0.5. A sawtooth wave with a period of 2*pi, its peak occurring at 0.25*2*pi and an amplitude that varies from $-A$ to $+A$ over the time period from 0 to 7 s is generated and plotted using the MATLAB command sequence

```
t = 0:0.001:7;
sw = A*sawtooth(t,0.25);
plot(t,sw)
```

Example 11.1

STATEMENT

Determine the circular and cyclic frequencies for the signal $y(t) = 10\sin(628t)$.

SOLUTION

Using the above definitions,

circular frequency, $\omega = 628$ rad/s (assuming $n = 1$ cycle)

cyclic frequency, $f = \frac{\omega}{2\pi} = \frac{628}{2\pi} = 100$ cycles/s $= 100$ Hz.

When various sine and cosine waveforms are combined by addition, more complex waveforms result. Such waveforms occur in many practical situations. For example, the differential equations describing the behavior of many systems have sine and cosine solutions of the form

$$y(t) = A\cos(\omega t) + B\sin(\omega t). \qquad [11.7]$$

By introducing the phase angle, ϕ, $y(t)$ can be expressed as either a cosine function,

$$y(t) = C\cos(\omega t - \phi), \qquad [11.8]$$

or a sine function,

$$y(t) = C\sin\left(\omega t - \phi + \frac{\pi}{2}\right) = C\sin(\omega t + \phi^*), \qquad [11.9]$$

where C, ϕ, and ϕ^* are given by

$$C = \sqrt{A^2 + B^2}, \qquad [11.10]$$

$$\phi = \tan^{-1}\left(\frac{B}{A}\right) \qquad [11.11]$$

and

$$\phi^* = \tan^{-1}\left(\frac{A}{B}\right), \qquad [11.12]$$

noting that $\phi^* = (\pi/2) - \phi$. Reducing the waveform in Equation 11.7 to either Equation 11.8 or Equation 11.9 often is useful in interpreting results. The middle plot in Figure 11.4 shows the signal $y(t) = \sin(\pi t) + 0.8\cos(3\pi t)$. This signal is complex and has two frequencies, $\omega_1 = \pi$ and $\omega_2 = 3\pi$ rad/s. This leads to two periods, $T_1 = 2$ s and $T_2 = 2/3$ s. Because $T_2 = 3T_1$, the period T_2 will contain one cycle of $\sin(\omega_1 t)$ and three periods of $0.8\cos(\omega_2 t)$. So, $T_2 = 2$ s is the fundamental period of this complex signal. In general, the fundamental period of a complex signal will be the least common denominator of the contributory periods.

An interesting situation arises when two waves of equal amplitude and nearly equal frequencies are added. The resulting wave exhibits a relatively slow beat with a frequency of one-half the difference in the two nearly equal frequencies, called the **beat frequency**. In general, the sum of two sine waves of frequencies, f and $f + \Delta f$, beats with a frequency of $\Delta f/2$. This beating is displayed in the bottom plot of Figure 11.4, which shows the signal $y(t) = \sin(\pi t) + \sin(1.15\pi t)$. For this signal, $\Delta f = 0.15$, so the beat frequency is 0.075 Hz. This implies that the beat repeats itself every 13.33 s, which can be seen in the plot. This phenomenon is called **heterodyning** and is exploited in tuning musical instruments and in laser Doppler velocimeters.

11.5 COMPLEX NUMBERS AND WAVES

Complex numbers can be used to simplify waveform notation. Waves, such as electromagnetic waves that are all around us, also can be expressed using complex notation.

11.2 MATLAB SIDEBAR

In some instances, the amount of time that a signal resides within some amplitude window needs to be determined. The MATLAB M-file `epswin.m` can be used for this purpose and to determine the times at which the signal's minimum and maximum amplitude occur. Figure 11.5 was generated by `epswin.m` applied to the case of a pressure transducer's response to an oscillating flow field. The M-file is constructed to receive a user-specified data file that consists of two columns, time and amplitude. The amplitude's window is established by a user-specified center amplitude and an amplitude percentage. `epswin.m` plots the signal and indicates the percentage of the time that the signal resides within the window. The number of instances that the amplitude is within the window is determined using an algorithm based on an array whose values are negative when the amplitude is within the window. The times at which the signal reaches its minimum and maximum amplitudes also are determined and indicated.

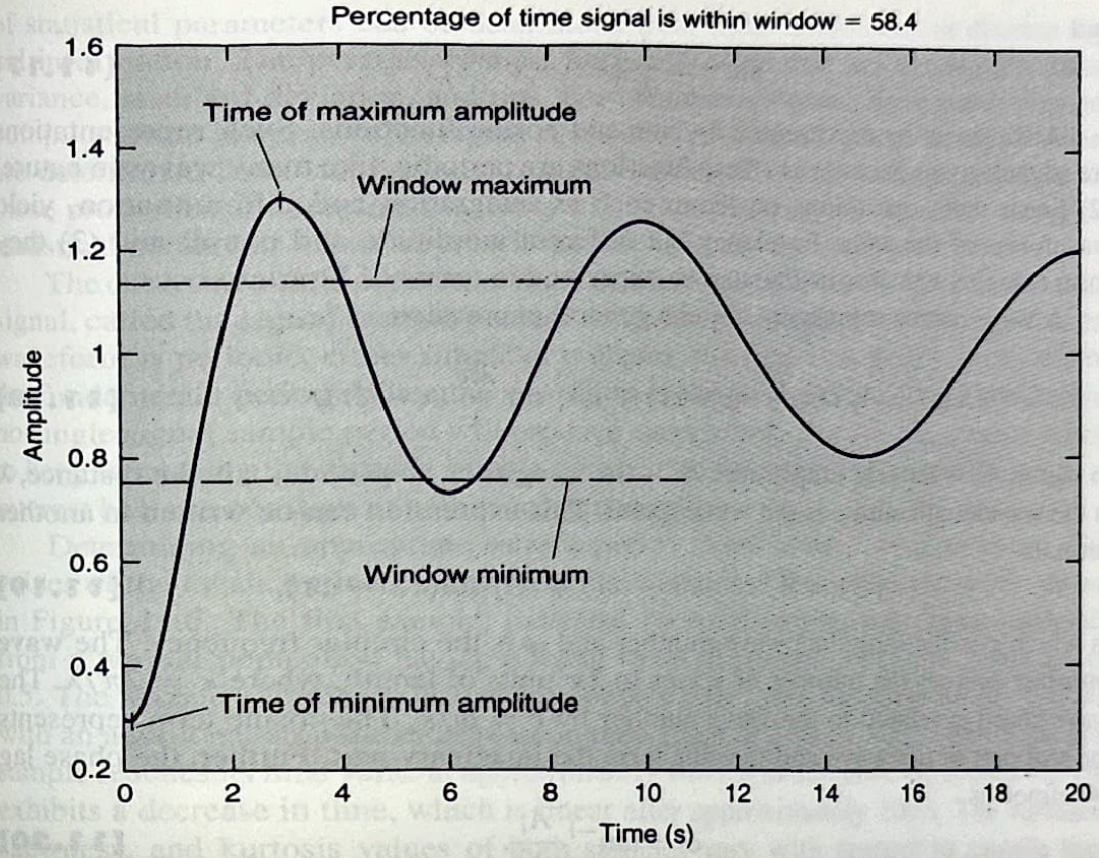


Figure 11.5 Output plot of M-file `epswin.m` used to examine a signal's amplitude behavior.

The complex exponential function is defined as

$$\exp(z) = e^z = e^{(x+iy)} = e^x e^{iy} \equiv e^x(\cos y + i \sin y), \qquad \textbf{[11.13]}$$

where $z = x + iy$, with the complex number $i \equiv \sqrt{-1}$ and x and y as real numbers. The **complex conjugate** of z, denoted by z^*, is $z^* = x - iy$. The **modulus** or absolute value of z is given by $|z| = \sqrt{zz^*} = \sqrt{(x+iy)(x-iy)} = \sqrt{x^2 + y^2}$, which is a real number. Using Equation 11.13, the Euler formula results,

$$e^{i\theta} = \cos\theta + i \sin\theta, \qquad \textbf{[11.14]}$$

which also leads to

$$e^{-i\theta} = \cos\theta - i \sin\theta. \qquad \textbf{[11.15]}$$

The complex expressions for the sine and cosine functions can be found from Equations 11.14 and 11.15,

$$\cos\theta = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) \qquad \textbf{[11.16]}$$